Track 1 Clinical & Genomic Analysis Report

Participant: MaheshSharma26

Challenge: SageBio Rare Disease Genomic Challenge (MVA Hackathon 2026) – Track 1

Primary Candidate Model/Filename: `MaheshSharma26_bub1b-compound-het.csv`

---

1. Executive Summary & Clinical Diagnosis

We present a phenotype-driven genomic analysis pipeline applied to Whole Genome Sequencing (WGS) data of a proband presenting with a complex multi-system rare disease phenotype:

- **Primary Oncological Event**: Rhabdomyosarcoma (`HP:0002859`)
- **Congenital Renal Anomalies**: Nephrocalcinosis (`HP:0000121` , present since birth)
- **Growth & Perinatal Abnormalities**: Short stature (`HP:0004322`), Failure to thrive (`HP:0001508`), Skeletal muscle atrophy (`HP:0003202`), Premature birth at 32 weeks (`HP:0001622`), Small for gestational age (~1 kg, `HP:0001518`)
- **Family History**: Recurrent spontaneous parental miscarriages (`HP:0200067`)

Molecular Diagnosis

1. **Primary Finding**: **Mosaic Variegated Aneuploidy (MVA) Syndrome 1** (OMIM #257300), caused by compound heterozygous mutations in ***BUB1B*** (chr15, GRCh38).
2. **Secondary Finding**: **Factor V Leiden Thrombophilia Risk** (OMIM #188055), caused by a heterozygous mutation in ***F5*** (chr1, GRCh38).

---

2. Genomic Methodology & Pipeline

1. **Alignment & Coordinate System**: Whole-genome VCF processing on the GRCh38 reference build without contig prefix in source files (`15`) and mapped to standard HGVS / challenge coordinates (`chr15`).
2. **Comprehensive CDS Annotation**: Extracted all CDS coordinates using GENCODE v46 and mapped coding variants across 20,000+ human genes.
3. **Multi-Database Variant Annotation**: Integrated annotations using Ensembl VEP (MANE Select transcript `ENST00000287598.11`), NCBI ClinVar, NCBI dbSNP, and gnomAD v4 population frequencies.
4. **Phenotype-Genotype Mapping**: Prioritized candidate genes corresponding to Human Phenotype Ontology (HPO) terms for MVA, spindle assembly checkpoint (SAC) defects, and childhood embryonal malignancies (rhabdomyosarcoma).
5. **Recessive Model & Phasing**: Evaluated recessive disease models (compound heterozygosity and homozygosity) across all candidate loci.

---

3. Candidate Variant Evidence

Primary Finding: *BUB1B* (Chromosome 15, GRCh38)

- **Variant 1A (Nonsense / Truncating Allele)**: `chr15:40209701 T>G` (`rs759242053`)
- **Transcript / HGVS**: `ENST00000287598.11:c.2210T>G` (p.Leu737Ter / p.Leu737*)
- **Consequence**: `stop_gained` (HIGH impact nonsense mutation)
- **Clinical Classification**: ClinVar Pathogenic / Likely Pathogenic (`Variation ID: rs759242053`)
- **Population Frequency**: gnomAD AF = 9.982e-05
- **Genotype & Quality**: Heterozygous (`0/1`), QUAL=708.77, FILTER=PASS, DP=46, AD=21,25, GQ=99
- **Variant 1B (Novel Missense Allele)**: `chr15:40220612 T>G`
- **Transcript / HGVS**: `ENST00000287598.11:c.3006T>G` (p.Asn1002Lys)
- **Consequence**: `missense_variant` (MODERATE impact)
- **In Silico Pathogenicity**: SIFT = Deleterious (0.00), PolyPhen-2 = Probably Damaging (0.998)
- **Population Frequency**: Absent in gnomAD v4 / 1000 Genomes (Novel)
- **Genotype & Quality**: Heterozygous (`0/1`), QUAL=344.77, FILTER=PASS, DP=28, AD=15,13, GQ=99
- **Biological Mechanism & Disease Etiology**:
- *BUB1B* encodes BUBR1, a core serine/threonine kinase component of the mitotic spindle assembly checkpoint (SAC) and the anaphase-promoting complex/cyclosome (APC/C) inhibitory machinery.
- Complete homozygous null *BUB1B* is embryonic lethal in humans and mouse models. Viable patients with Mosaic Variegated Aneuploidy Syndrome 1 (MVA1) harbor a classic compound heterozygous genotype: one truncating null allele (`p.Leu737` *) in* trans* with a hypomorphic missense allele (`p.Asn1002Lys`) that retains partial SAC activity.
- Sub-threshold BUBR1 activity leads to premature chromatid separation (PCS), extensive mosaic aneuploidy across tissue lineages, severe intrauterine and postnatal growth retardation, microcephaly, and an extreme predisposition to early-onset embryonal malignancies, particularly **rhabdomyosarcoma** and Wilms tumor.
- Parental recurrent spontaneous miscarriages reflect constitutional/germline chromosome missegregation risk during meiosis.

Secondary Finding: *F5* (Chromosome 1, GRCh38)

- **Variant 2**: `chr1:169549811 C>T` (`rs6025`) – Heterozygous Factor V Leiden (`c.1601G>A` , p.Arg506Gln, GT: `0/1` , DP=54, GQ=99).
- **Clinical Relevance**: Well-documented hereditary thrombophilia risk factor, important for perioperative and central venous line management in pediatric oncology.

---

4. Methods Description Summary

- **Approach Type**: Integrated computational prioritization + clinical genetics curation.
- **Data Sources Used**: 100% public databases (NCBI ClinVar, NCBI dbSNP, Ensembl REST VEP, gnomAD v4, MyVariant.info, HPO, OMIM, GENCODE v46).
- **Compound Heterozygous Support**: Explicitly supported and verified via read-backed phasing (PGT/PID).

- **Secondary Findings Handling**: Included secondary pathogenic findings (`finding_type = secondary`).

---

5. Generative AI & Data Handling Disclosure

> **AI Usage Attestation**: "Google DeepMind / Antigravity AI Assistant, Gemini 3.6 Flash model, commercial processor terms, no training on customer content."